chemistry

1)

a) Magnesium + oxygen → Magnesium oxide.

$2Mg + O_2 \rightarrow 2MgO$

b) Iron + copper sulphate → Iron sulphate + copper

$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

3)

The calorific value of a fuel is the amount of heat produced when one kilogram of the fuel is completely burnt. SI unit : KJ/Kg

4) A displacement reaction is a reaction in which a more reactive element displaces a less reactive element from its compound. when a copper strip is dipped into zinc sulphate ($ZnSO_4$) solution, no reaction occurs

6) Destructive distillation of coal is the process of heating coal strongly at a high temperature in the absence of air.

→ coke
→ coal tar.

7)

a) Metals generally react with dilute acids to form salt and hydrogen gas.

eg: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

b) The gas produced is hydrogen (H_2).

Test: Bring a burning splint near the gas. Hydrogen burns with a characteristic 'Pop' sound.

8) A candle flame has three zones:

1) outer zone: blue / non-luminous; complete combustion; hottest zone.

2) Middle zone: yellow / luminous; Partial combustion

3) Inner zone: dark zone; Contains unburnt wax vapours; Coolest zone;

10) a) Plastics are widely used because they are

1) lightweight
2) Strong and durable
3) water-resistant

c) 5R Principle

Refuse → Reduce → Reuse → Repurpose - Recycle.